Louis Desgree Implementing a Linux Router in UTM Kali Linux

Introduction

This report outlines the implementation of a Linux router within a virtualized environment using Kali Linux in UTM. The objective was to configure and test a router capable of forwarding network traffic between two networks. This process involved setting up network interfaces with static IP addresses, enabling IP forwarding, configuring Network Address Translation (NAT), and verifying connectivity. Screenshots were captured throughout the process to validate the implementation and document the configuration.

System Setup

To establish the routing environment, the existing Kali Linux virtual machine within UTM was configured with two network interfaces. The first interface, enp0s1, was designated as the external network interface and assigned a static IP address of **192.168.64.5** with a subnet mask of **255.255.255.0**. The second interface, enp0s2, was configured as the internal network interface with a static IP address of **10.37.129.4** and the same subnet mask. This internal network serves as the private network that routes traffic through the Kali Linux machine.

Configuring these interfaces required modifying the system's network configuration file at **/etc/network/interfaces**:

```
auto lo
iface lo inet loopback

auto enp0s1
iface enp0s1 inet static
   address 192.168.64.5
   netmask 255.255.255.0

auto enp0s2
iface enp0s2 inet static
   address 10.37.129.4
   netmask 255.255.255.0
```

After updating the file, the network service was restarted using:

```
sudo ifdown enp0s1 && sudo ifup enp0s1
sudo ifdown enp0s2 && sudo ifup enp0s2
```

This ensured that both interfaces used the correct static configurations.

Enabling IP Forwarding

For the Kali Linux machine to function as a router, IP forwarding had to be enabled. This was temporarily achieved by:

```
echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward
```

To ensure persistence across reboots, the system configuration file /etc/sysctl.conf was edited by adding or uncommenting the line:

```
net.ipv4.ip_forward = 1
```

Changes were applied with:

```
sudo sysctl -p
```

Configuring Routing and NAT

With IP forwarding enabled, NAT was configured to allow internal clients to access external networks. This was accomplished using **iptables**:

```
sudo iptables -t nat -A POSTROUTING -o enp0s1 -j MASQUERADE
sudo iptables -A FORWARD -i enp0s2 -o enp0s1 -j ACCEPT
sudo iptables -A FORWARD -i enp0s1 -o enp0s2 -m state --state RELATED,ESTABLISHED
-j ACCEPT
```

To persist these rules across reboots, the **iptables-persistent** package was installed, and the rules were saved:

```
sudo netfilter-persistent save
```

Testing the Router

After completing the configuration, multiple tests were conducted to verify functionality:

1. **Ping Test (Internal Network)**: From an internal client, a ping was sent to **10.37.129.4** (internal interface of the router) to verify internal connectivity.

IP Forwarding Check: The value of **/proc/sys/net/ipv4/ip_forward** was checked to ensure forwarding was enabled:

```
cat /proc/sys/net/ipv4/ip_forward
```

2. The output should be **1**.

Routing Table Validation: To verify the correct routes were established:
ip route show

3.

External Connectivity Test: From an internal client, a ping was sent to **8.8.8.8** to test internet access:
ping 8.8.8.8

4. A successful response confirmed proper NAT and routing.

Conclusion

This report documented the process of implementing a Linux router using Kali Linux within UTM. The setup included configuring network interfaces with static IPs, enabling IP forwarding, establishing NAT rules, and verifying the router's functionality. Each step was carefully tested, and screenshots were captured to demonstrate the results. The successful completion of this project provided a functional router capable of forwarding packets between two networks while allowing internet access through NAT. This implementation aligns with the requirements outlined in Chapter 4 of *Building Virtual Pentesting Labs for Advanced Penetration Testing* and serves as a practical exercise in network infrastructure management.

References

https://www.tecmint.com/setup-linux-as-router/

Screenshots

```
root@-Standard-PC-Q35-ICH9-2009:~# cat /etc/network/interfaceses
# interfaces(5) file used by ifup(8) and ifdown(8)
auto lo
iface lo inet loopback

auto enp0s1
iface enp0s1 inet static
address 192.168.64.5
netmask 255.255.255.0

auto enp0s2
iface enp0s2 inet static
address 10.37.129.4
netmask 255.255.255.0
```

```
root@Standard-PC-Q35-ICH9-2009:~# cat /proc/sys/net/ipv4/ip_forward
1
```